Data Sufficiency Analysis

Executive Summary

Current dataset size: 2,880 rows across four regions and nine Dynamic World classes.

- S1_VV: best observed learning-curve result is Lightgbm with Embeddings + region + Dynamic World at validation R2=0.902; assessment: Curve is near plateau.

- S1_VH: best observed learning-curve result is Lightgbm with Embeddings + region + Dynamic World at validation R2=0.943; assessment: Curve is near plateau.

- S1_VV_div_VH: best observed learning-curve result is Ridge with Embeddings + region + Dynamic World at validation R2=0.708; assessment: Curve is near plateau.

- Spatial-block effective sample size is about 57 out of 2,880 nominal rows, so the dataset is materially clustered.

- Mean nearest-neighbor embedding cosine similarity is 0.955, which indicates moderate redundancy but not collapse into near-duplicates.

Recommendation

This analysis asks whether the current 2,880-row dataset is large enough to support stable conclusions for the three main SAR targets.

- S1_VV: best curve uses Embeddings + region + Dynamic World with Lightgbm; held-out R2 at full size is 0.902, tail gain from 80% to 100% is 0.007, CV std is 0.012, region spread is 0.097, class spread is 0.218. More data is not the primary bottleneck.

Data Sufficiency Analysis (cont.)

- S1_VH: best curve uses Embeddings + region + Dynamic World with Lightgbm; held-out R2 at full size is 0.943, tail gain from 80% to 100% is 0.004, CV std is 0.004, region spread is 0.081, class spread is 0.134. More data is not the primary bottleneck.

- S1_VV_div_VH: best curve uses Embeddings + region + Dynamic World with Ridge; held-out R2 at full size is 0.708, tail gain from 80% to 100% is 0.003, CV std is 0.027, region spread is 0.427, class spread is 0.367. More generic rows are unlikely to change the result much; target quality or features are the tighter bottleneck.

- Spatial clustering reduces the nominal sample count. The 2,880 rows occupy 64 spatial blocks, with a cluster-size effective sample size of about 57.

- Region x class balancing is stronger than raw spatial independence: the region/class effective sample size is about 36.

- Embedding redundancy is moderate rather than extreme: mean nearest-neighbor cosine similarity is 0.955 and the 95th percentile is 0.998.

VV and VH look closer to a feature-limited or target-limited regime than a purely data-limited regime.

The polarization-difference target is weaker, but the remaining gap may be feature-driven as much as sample-driven.

If more data is added, the highest-value additions are more independent spatial samples, more diverse regions, and targeted examples from difficult land-cover conditions and extreme polarization-difference cases rather than simply duplicating the current balanced design.

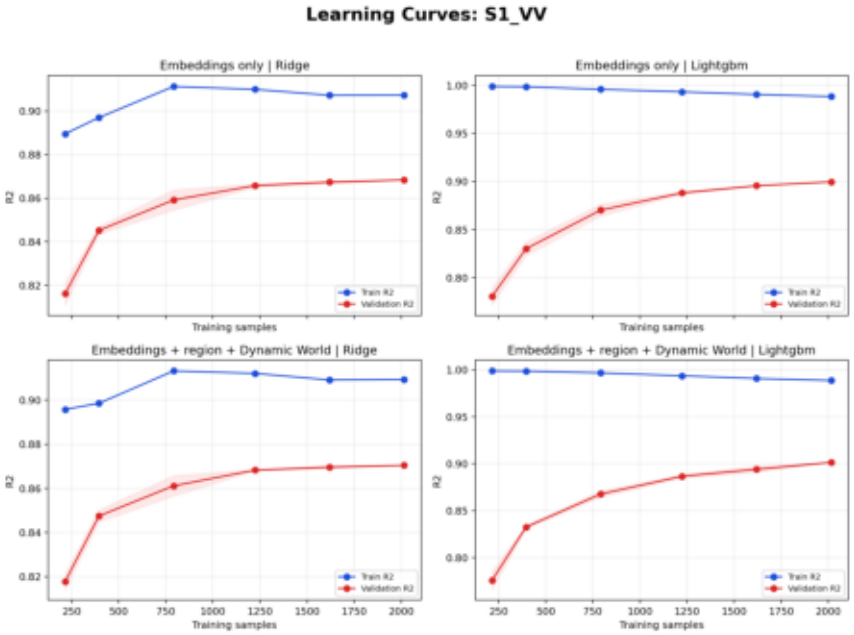
Learning Curve Summary

target	best_feature_set	best_model	best_valid_r2	tail_gain_80_to_100	best_cv_r2_std	assessment
S1_VV	Embeddings + region + Dynamic World	Lightgbm	0.902	0.007	0.012	Curve is near plateau
S1_VH	Embeddings + region + Dynamic World	Lightgbm	0.943	0.004	0.004	Curve is near plateau
S1_VV_div_V	Embeddings + region + Dynamic World	Ridge	0.708	0.003	0.027	Curve is near plateau

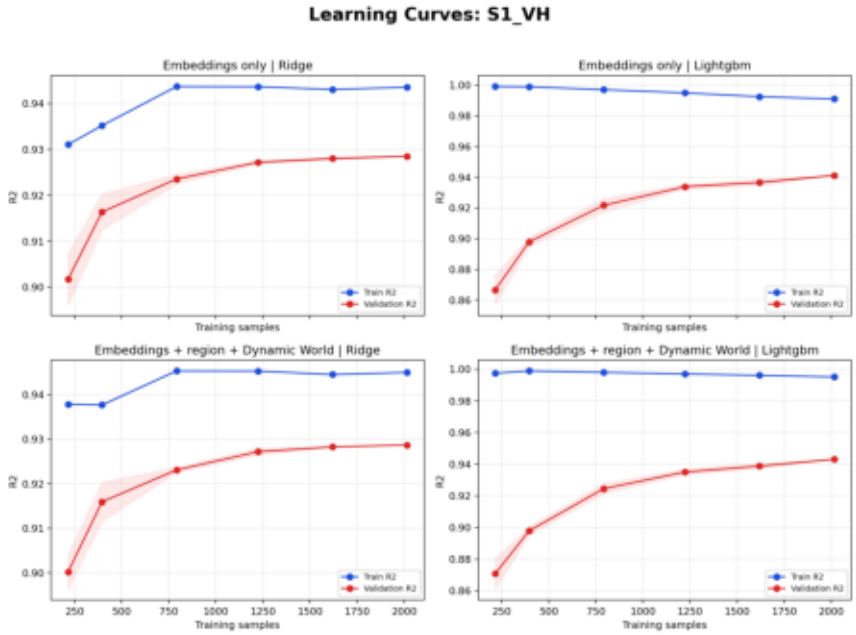
Assessment is based on the validation learning curve tail from 80% to 100% of the current training set and the repeated grouped-CV variance of the best-performing model/feature combination.

Learning Curves

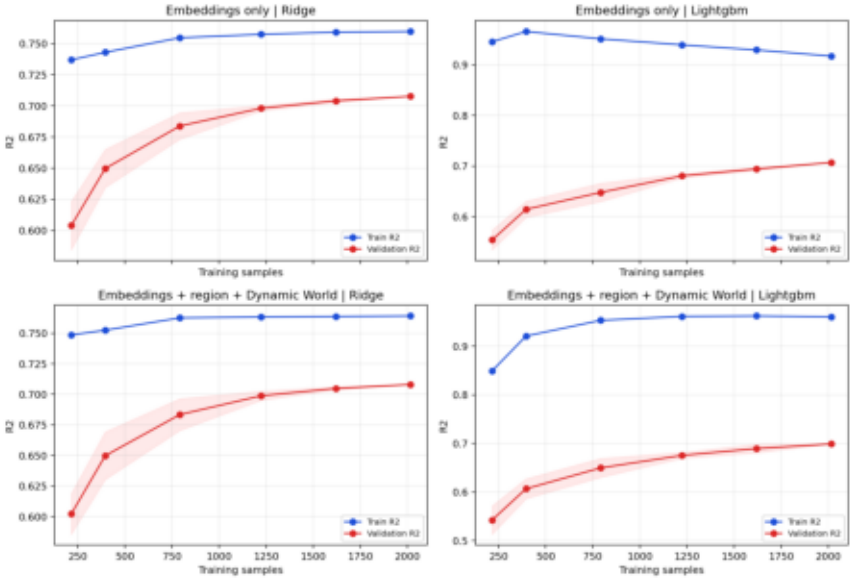
data sufficiency learning curve S1 VV



data sufficiency learning curve S1 VH

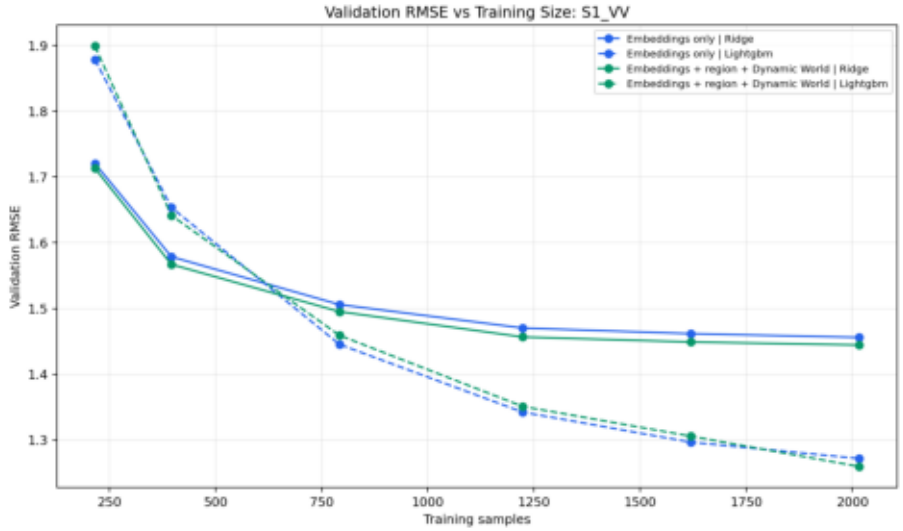


data sufficiency learning curve S1 VV div VH

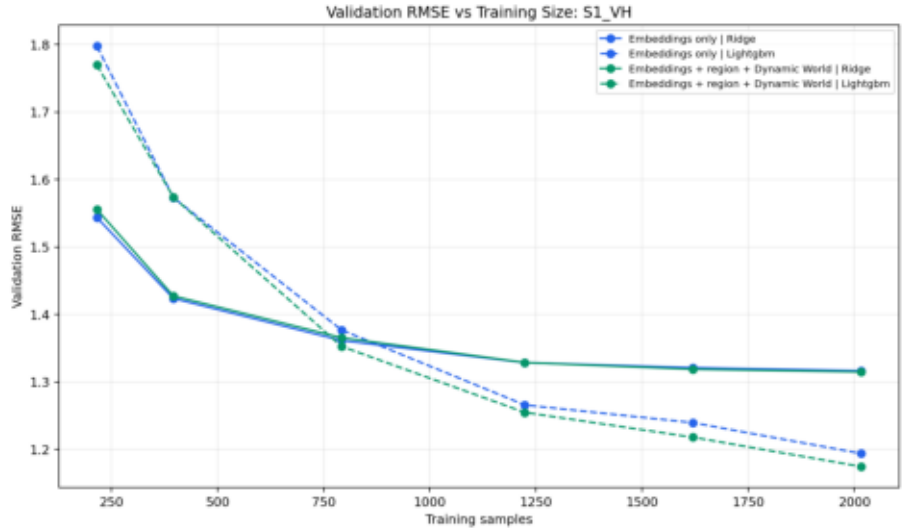


Learning Curve RMSE

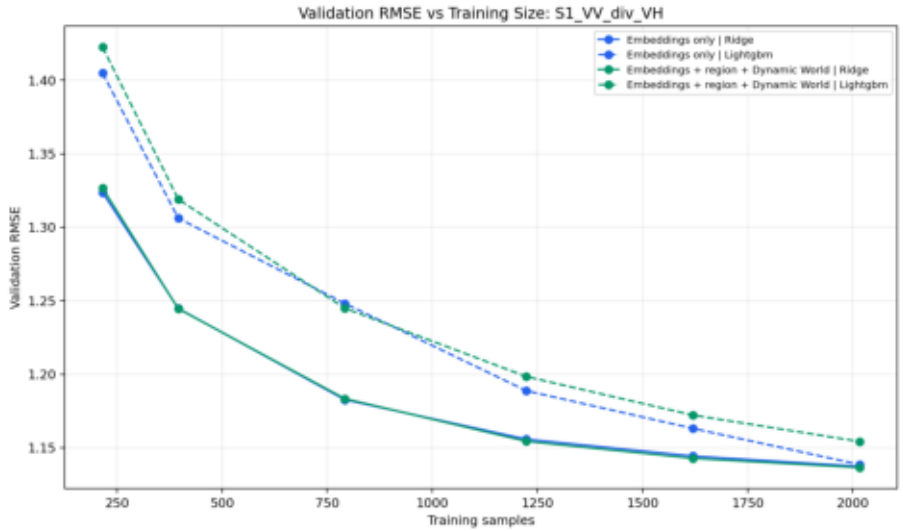
data sufficiency learning curve rmse S1 VV



data sufficiency learning curve rmse S1 VH



data sufficiency learning curve rmse S1 VV div VH



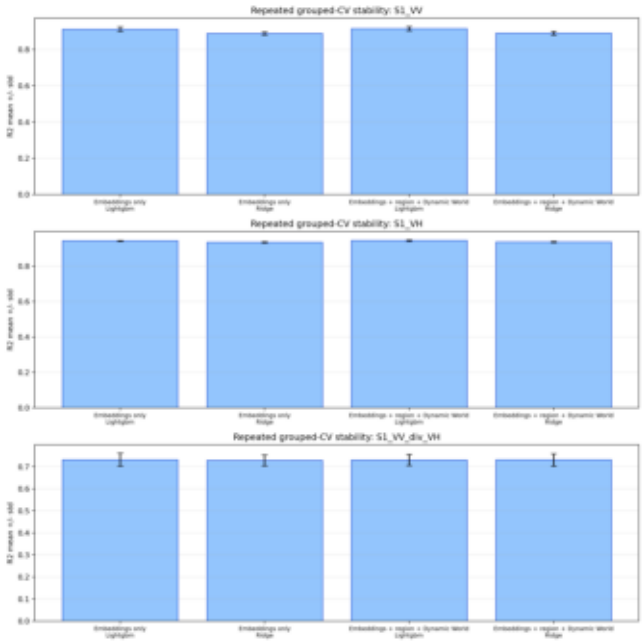
Repeated Grouped-CV Stability

feature_set	target	model_name	r2_mean	r2_std	rmse_mean	rmse_std	mae_mean	mae_std	pearson_r_mean	folds
Embeddings only	S1_VH	Lightgbm	0.943	0.002	1.180	0.027	0.876	0.008	0.972	12
Embeddings only	S1_VH	Ridge	0.936	0.003	1.259	0.027	0.935	0.040	0.967	12
Embeddings only	S1_VV	Lightgbm	0.913	0.012	1.144	0.076	0.813	0.030	0.956	12
Embeddings only	S1_VV	Ridge	0.888	0.009	1.294	0.054	0.909	0.026	0.943	12
Embeddings only	S1_VV_div_VH	Lightgbm	0.732	0.029	1.063	0.049	0.786	0.025	0.856	12
Embeddings only	S1_VV_div_VH	Ridge	0.729	0.025	1.070	0.036	0.787	0.024	0.855	12
Embeddings + region + Dynamic World	S1_VH	Lightgbm	0.945	0.004	1.168	0.042	0.876	0.026	0.972	12
Embeddings + region + Dynamic World	S1_VH	Ridge	0.937	0.004	1.248	0.035	0.922	0.043	0.968	12
Embeddings + region + Dynamic World	S1_VV	Lightgbm	0.915	0.012	1.130	0.075	0.805	0.032	0.957	12
Embeddings + region + Dynamic World	S1_VV	Ridge	0.890	0.010	1.284	0.063	0.894	0.037	0.944	12
Embeddings + region + Dynamic World	S1_VV_div_VH	Lightgbm	0.731	0.025	1.066	0.046	0.788	0.023	0.855	12
Embeddings + region + Dynamic World	S1_VV_div_VH	Ridge	0.731	0.027	1.065	0.036	0.783	0.029	0.856	12

Each row aggregates 12 validation folds per setup: 3 random seeds x 4 grouped folds.

Cross-Validation Stability

data sufficiency cv stability r2



Coverage By Region

region	n_samples
amazon_forest	720
california_coast	720
iowa_ag	720
sf_bay_urban	720

Coverage By Dynamic World Class

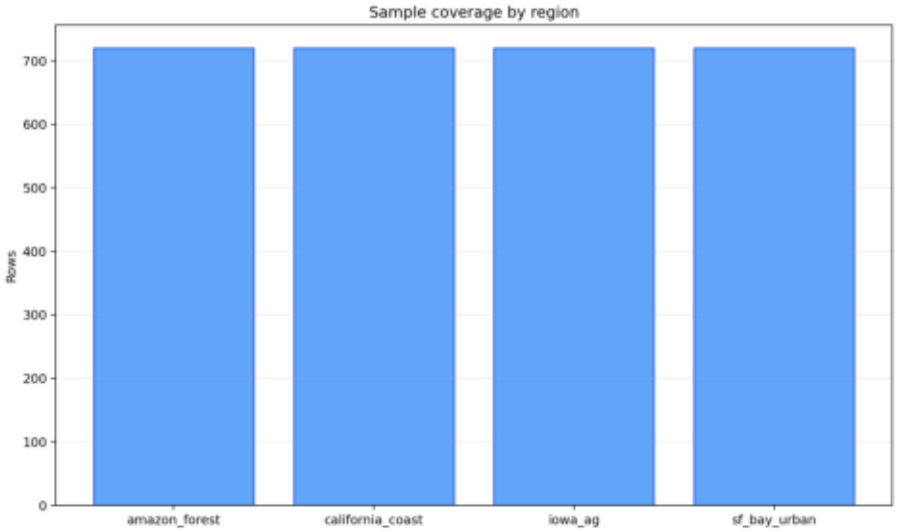
dw_label_name	n_samples
bare	320
built	320
crops	320
flooded_vegetation	320
grass	320
shrub_and_scrub	320
snow_and_ice	320
trees	320
water	320

Coverage By Region x Class

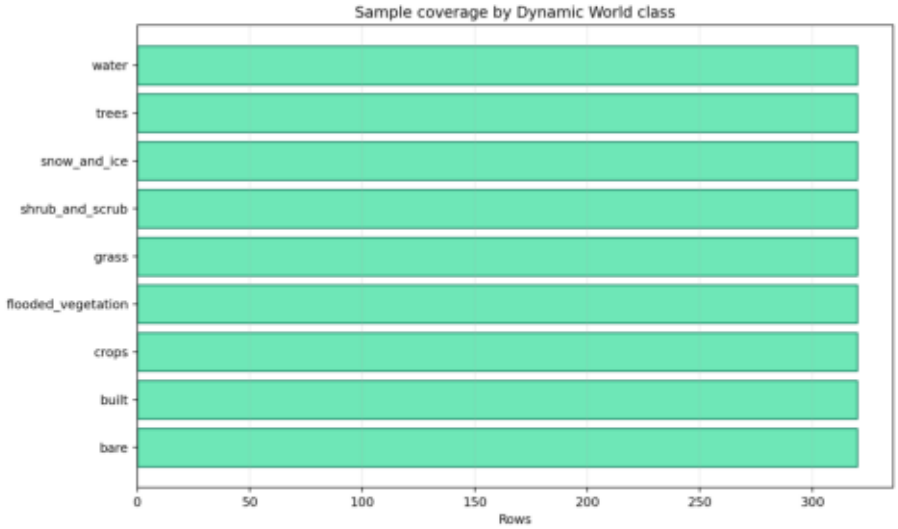
region	dw_label_name	n_samples
amazon_forest	bare	80
amazon_forest	built	80
amazon_forest	crops	80
amazon_forest	flooded_vegetation	80
amazon_forest	grass	80
amazon_forest	shrub_and_scrub	80
amazon_forest	snow_and_ice	80
amazon_forest	trees	80
amazon_forest	water	80
california_coast	bare	80
california_coast	built	80
california_coast	crops	80
california_coast	flooded_vegetation	80
california_coast	grass	80
california_coast	shrub_and_scrub	80
california_coast	snow_and_ice	80
california_coast	trees	80
california_coast	water	80
iowa_ag	bare	80
iowa_ag	built	80
iowa_ag	crops	80
iowa_ag	flooded_vegetation	80
iowa_ag	grass	80
iowa_ag	shrub_and_scrub	80
iowa_ag	snow_and_ice	80
iowa_ag	trees	80
iowa_ag	water	80
sf_bay_urban	bare	80
sf_bay_urban	built	80
sf_bay_urban	crops	80
sf_bay_urban	flooded_vegetation	80
sf_bay_urban	grass	80
sf_bay_urban	shrub_and_scrub	80
sf_bay_urban	snow_and_ice	80
sf_bay_urban	trees	80
sf_bay_urban	water	80

Coverage Visuals

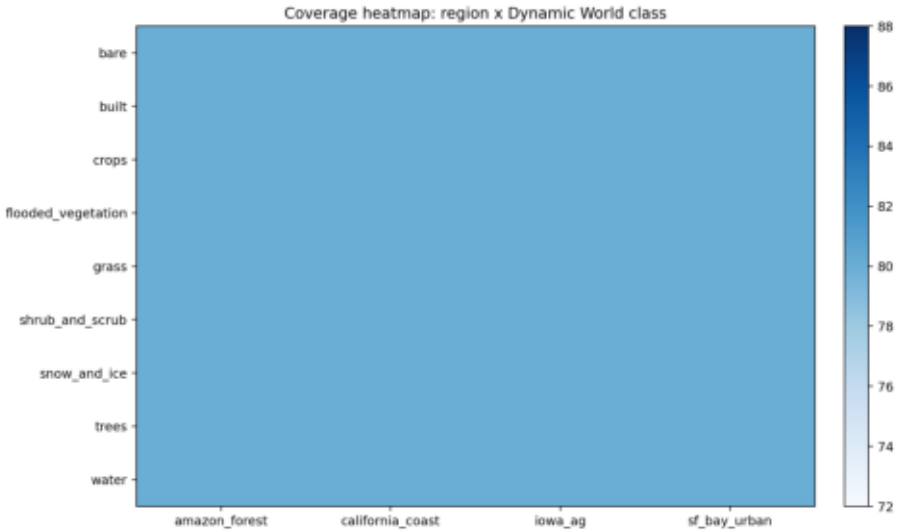
data sufficiency coverage by region



data sufficiency coverage by land cover



data sufficiency coverage heatmap region land cover



Best Model Per Target

target	feature_set	model_nam	r2_mean	r2_std	rmse_mea	rmse_std	mae_mea	mae_std	pearson_r_mean	folds
S1_VH	Embeddings + region + Dynamic World	Lightgbm	0.945	0.004	1.168	0.04171375980697468	0.876	0.025696468307712498	0.9721127388215001	12
S1_VV	Embeddings + region + Dynamic World	Lightgbm	0.915	0.012	1.130	0.07474214226418306	0.805	0.032240843246580816	0.9570594777479102	12
S1_VV_div_V	Embeddings only	Lightgbm	0.732	0.029	1.063	0.04882782023706216	0.786	0.024753113647795484	0.856282624055699	12

Best Model Region Performance

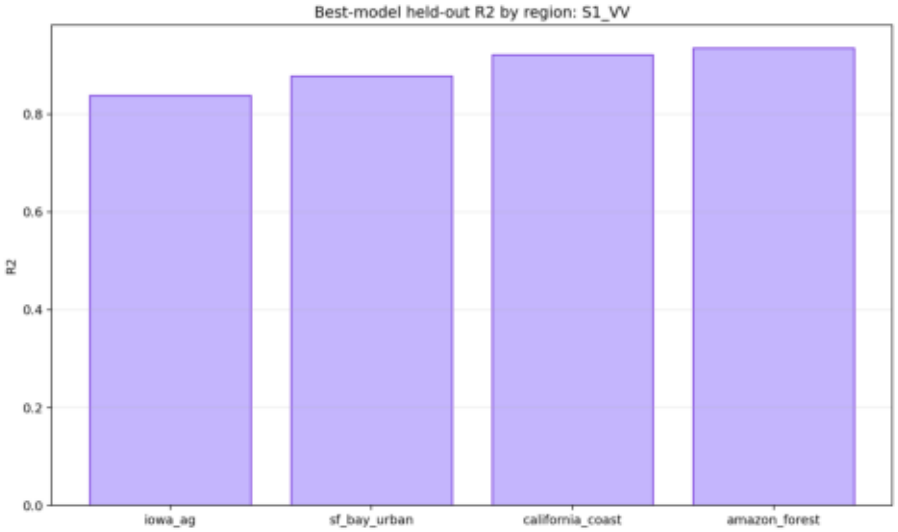
target	feature_set	model_name	region	n_samples	r2	rmse	mae	pearson_r
S1_VH	Embeddings + region + Dynamic World	Lightgbm	iowa_ag	216	0.878	1.173	0.880	0.938
S1_VH	Embeddings + region + Dynamic World	Lightgbm	sf_bay_urban	216	0.933	1.369	0.991	0.966
S1_VH	Embeddings + region + Dynamic World	Lightgbm	amazon_forest	216	0.951	1.103	0.812	0.976
S1_VH	Embeddings + region + Dynamic World	Lightgbm	california_coast	216	0.959	1.027	0.748	0.980
S1_VV	Embeddings + region + Dynamic World	Lightgbm	iowa_ag	216	0.839	1.357	0.789	0.923
S1_VV	Embeddings + region + Dynamic World	Lightgbm	sf_bay_urban	216	0.878	1.474	0.981	0.940
S1_VV	Embeddings + region + Dynamic World	Lightgbm	california_coast	216	0.922	1.029	0.750	0.961
S1_VV	Embeddings + region + Dynamic World	Lightgbm	amazon_forest	216	0.935	1.129	0.778	0.967
S1_VV_div_VH	Embeddings only	Lightgbm	iowa_ag	216	0.416	1.366	0.860	0.669
S1_VV_div_VH	Embeddings only	Lightgbm	amazon_forest	216	0.484	1.042	0.746	0.704
S1_VV_div_VH	Embeddings only	Lightgbm	sf_bay_urban	216	0.726	1.123	0.820	0.852
S1_VV_div_VH	Embeddings only	Lightgbm	california_coast	216	0.843	0.987	0.759	0.918

Best Model Land-Cover Performance

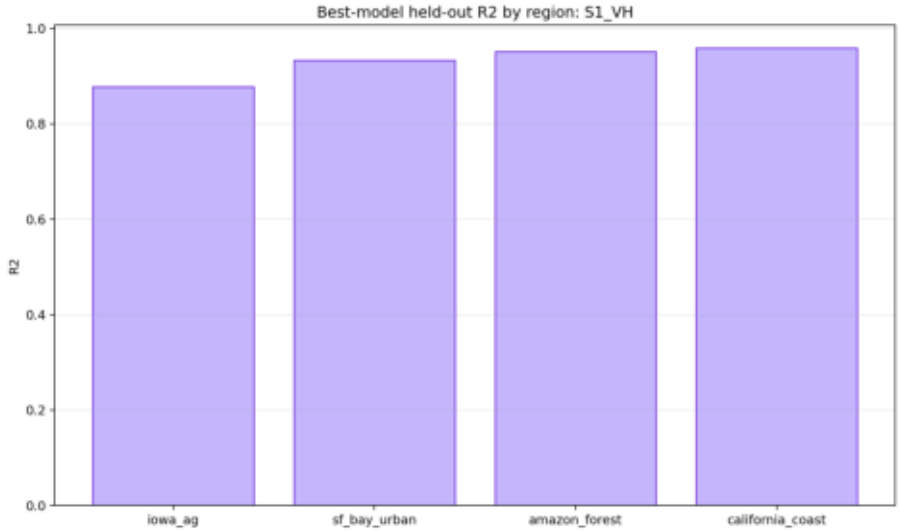
target	feature_set	model_name	dw_label_name	n_samples	r2	rmse	mae	pearson_r
S1_VH	Embeddings + region + Dynamic World	Lightgbm	bare	96	0.792	1.075	0.775	0.891
S1_VH	Embeddings + region + Dynamic World	Lightgbm	shrub_and_scrub	96	0.812	1.397	0.884	0.904
S1_VH	Embeddings + region + Dynamic World	Lightgbm	trees	96	0.843	0.990	0.784	0.920
S1_VH	Embeddings + region + Dynamic World	Lightgbm	built	96	0.857	1.304	0.935	0.932
S1_VH	Embeddings + region + Dynamic World	Lightgbm	snow_and_ice	96	0.873	1.197	0.926	0.936
S1_VH	Embeddings + region + Dynamic World	Lightgbm	water	96	0.891	1.580	1.230	0.945
S1_VH	Embeddings + region + Dynamic World	Lightgbm	crops	96	0.896	0.858	0.704	0.950
S1_VH	Embeddings + region + Dynamic World	Lightgbm	grass	96	0.896	1.060	0.789	0.947
S1_VH	Embeddings + region + Dynamic World	Lightgbm	flooded_vegetation	96	0.925	0.918	0.693	0.963
S1_VV	Embeddings + region + Dynamic World	Lightgbm	bare	96	0.699	1.189	0.826	0.849
S1_VV	Embeddings + region + Dynamic World	Lightgbm	built	96	0.732	2.107	1.233	0.872
S1_VV	Embeddings + region + Dynamic World	Lightgbm	crops	96	0.782	0.978	0.724	0.885
S1_VV	Embeddings + region + Dynamic World	Lightgbm	shrub_and_scrub	96	0.788	1.345	0.894	0.893
S1_VV	Embeddings + region + Dynamic World	Lightgbm	trees	96	0.847	0.938	0.744	0.923
S1_VV	Embeddings + region + Dynamic World	Lightgbm	water	96	0.850	1.462	0.988	0.931
S1_VV	Embeddings + region + Dynamic World	Lightgbm	grass	96	0.858	0.978	0.691	0.929
S1_VV	Embeddings + region + Dynamic World	Lightgbm	snow_and_ice	96	0.898	1.016	0.761	0.948
S1_VV	Embeddings + region + Dynamic World	Lightgbm	flooded_vegetation	96	0.917	0.810	0.559	0.958
S1_VV_div_VH	Embeddings only	Lightgbm	trees	96	0.358	0.600	0.483	0.642
S1_VV_div_VH	Embeddings only	Lightgbm	shrub_and_scrub	96	0.378	0.932	0.757	0.635
S1_VV_div_VH	Embeddings only	Lightgbm	built	96	0.463	1.846	1.126	0.712
S1_VV_div_VH	Embeddings only	Lightgbm	crops	96	0.532	1.000	0.782	0.734
S1_VV_div_VH	Embeddings only	Lightgbm	bare	96	0.547	1.047	0.780	0.748
S1_VV_div_VH	Embeddings only	Lightgbm	grass	96	0.647	0.956	0.708	0.810
S1_VV_div_VH	Embeddings only	Lightgbm	snow_and_ice	96	0.661	1.034	0.760	0.820
S1_VV_div_VH	Embeddings only	Lightgbm	flooded_vegetation	96	0.663	0.822	0.618	0.832
S1_VV_div_VH	Embeddings only	Lightgbm	water	96	0.724	1.509	1.152	0.854

Subgroup Stability

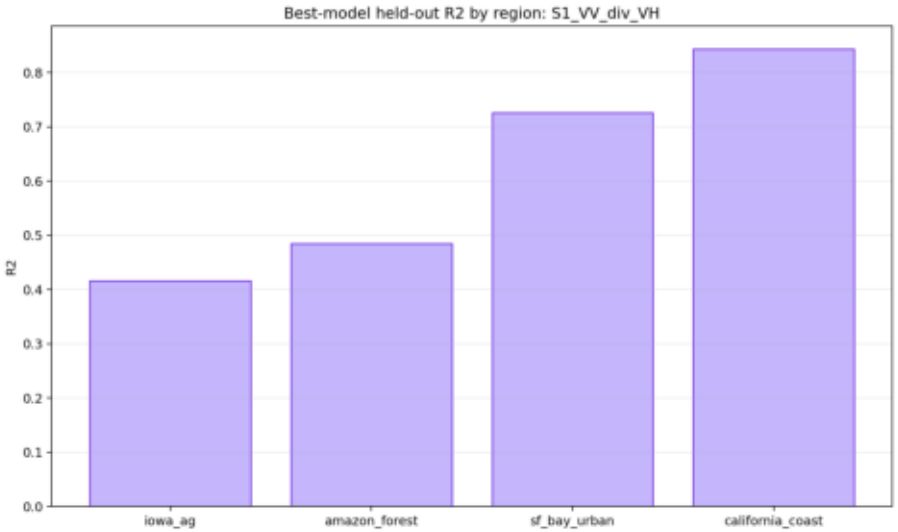
data sufficiency subgroup region r2 S1 VV



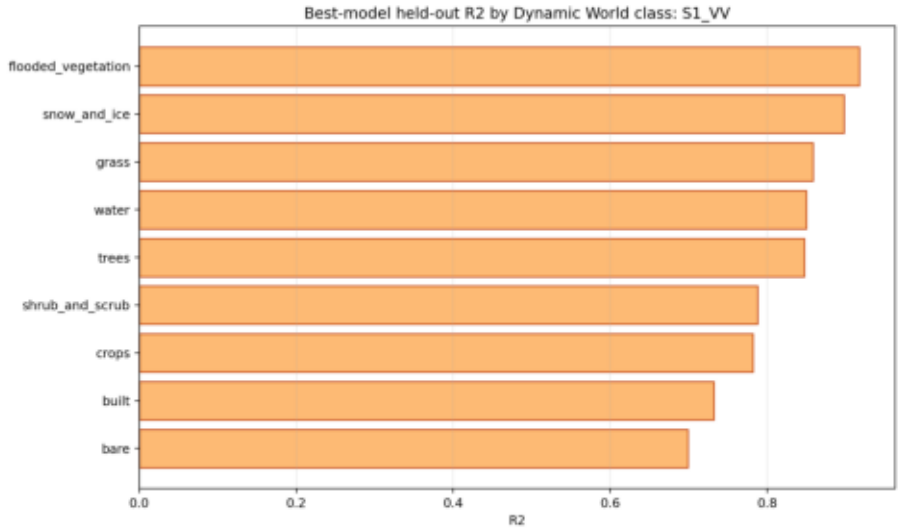
data sufficiency subgroup region r2 S1 VH



data sufficiency subgroup region r2 S1 VV div VH

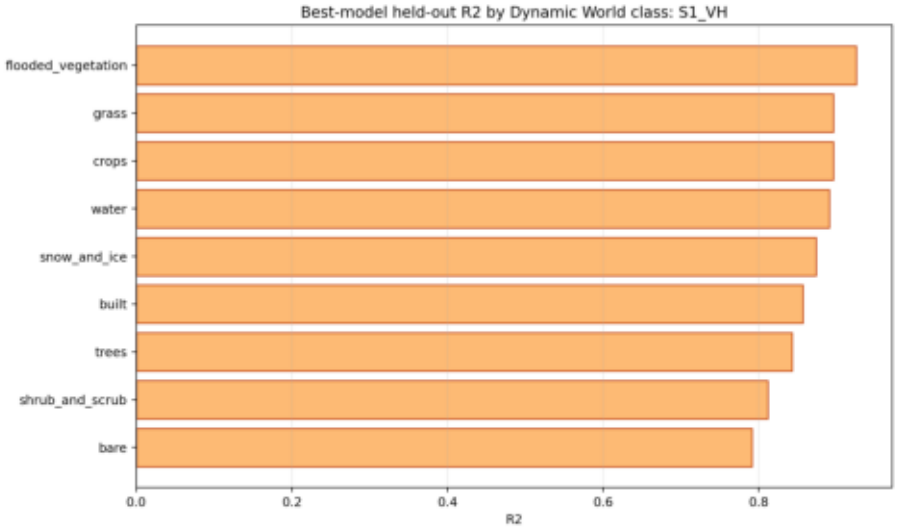


data sufficiency subgroup land cover r2 S1 VV

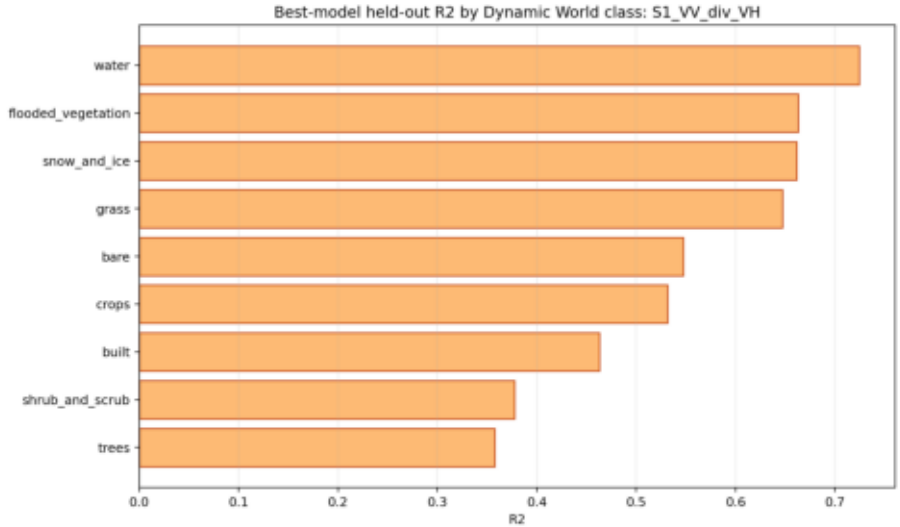


Subgroup Stability (cont.)

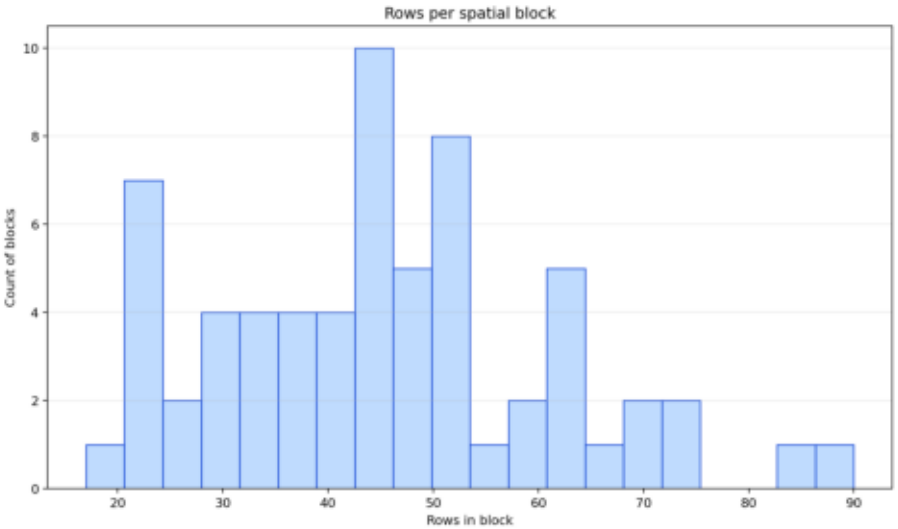
data sufficiency subgroup land cover r2 S1 VH



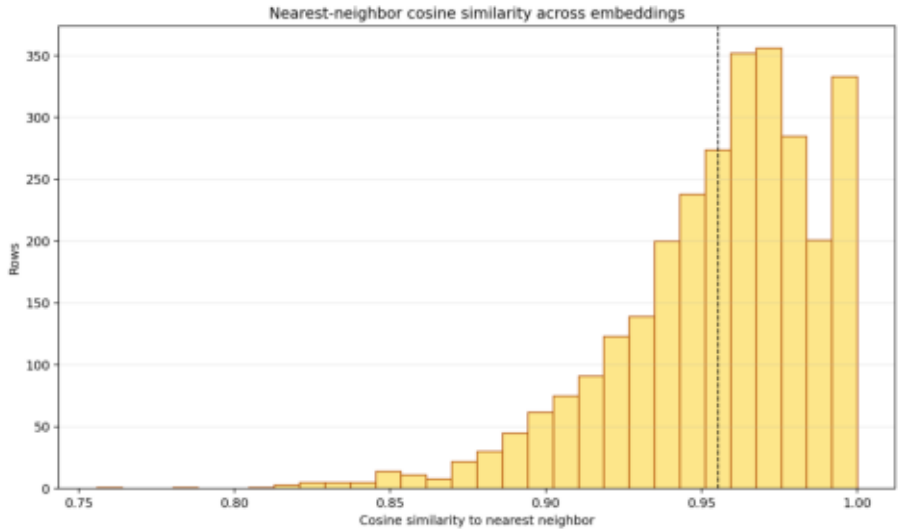
data sufficiency subgroup land cover r2 S1 VV div VH



data sufficiency redundancy spatial block histogram



data sufficiency redundancy nearest neighbor cosine



Redundancy Diagnostics

- Spatial blocks: 64 total, mean 45.0 rows, max 90 rows.

- Spatial-block effective sample size: 56.9.

- Region/class effective sample size: 36.0.

- Mean nearest-neighbor cosine: 0.955.

- 95th percentile nearest-neighbor cosine: 0.998.

- Exact duplicate embedding rows: 5.

This analysis asks whether the current 2,880-row dataset is large enough to support stable conclusions for the three main SAR targets.

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- Spatial clustering reduces the nominal sample count. The 2,880 rows occupy 64 spatial blocks, with a cluster-size effective sample size of about 57.

Redundancy Diagnostics (cont.)

- Region x class balancing is stronger than raw spatial independence: the region/class effective sample size is about 36.

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VV and VH look closer to a feature-limited or target-limited regime than a purely data-limited regime.

The polarization-difference target is weaker, but the remaining gap may be feature-driven as much as sample-driven.

If more data is added, the highest-value additions are more independent spatial samples, more diverse regions, and targeted examples from difficult land-cover conditions and extreme polarization-difference cases rather than simply duplicating the current balanced design.